Lost & Found Inside Train Coach

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PROBLEM STATEMENT

Passengers lose items inside moving trains.

No precise location tracking

- **Train keeps moving → difficult to identify where item was lost**
- **No coach-level location data**
- **No real-time map visualization**

PROPOSED SMART SOLUTION

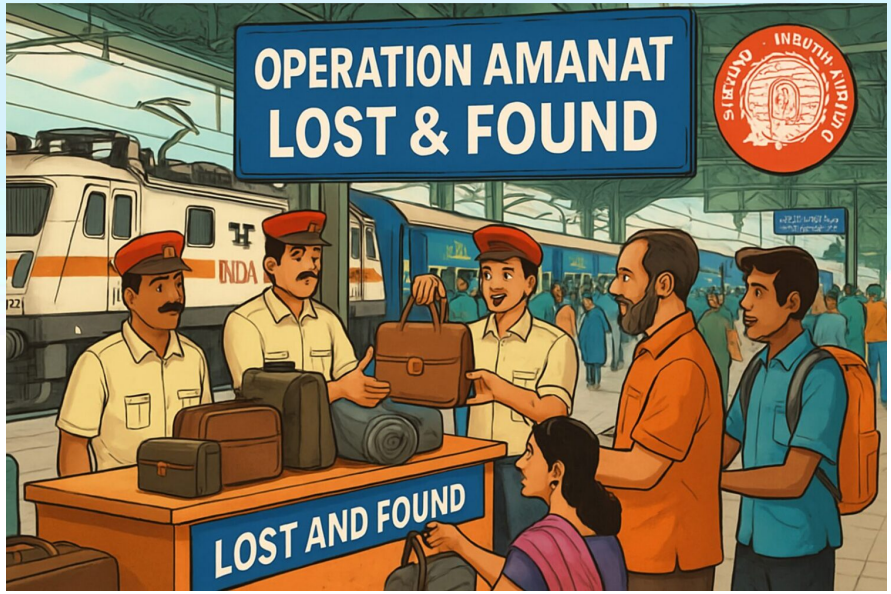
Each coach has:

- Low-cost GPS module
- QR code for reporting
- Offline map stored in app

When passenger reports item:

System captures:

- Coach number
- Seat number
- Real-time GPS coordinates
- Timestamp
- Location stored in database
- Even if internet disconnects → offline data saved & synced later



AS-IS VS TO-BE PROCESS MAP

Feature	As-Is (Current)	To-Be (Proposed)
Reporting	Physical visit or phone calls.	Instant report via App/QR code on seat.
Tracking	Manual ledger entries/Paperwork.	Centralized Cloud Database.
Identification	Vague descriptions.	Image-based AI matching.
Return	Must return to the specific station.	Option for "Courier-to-Home" (Paid).

OFFLINE MAP CONCEPT

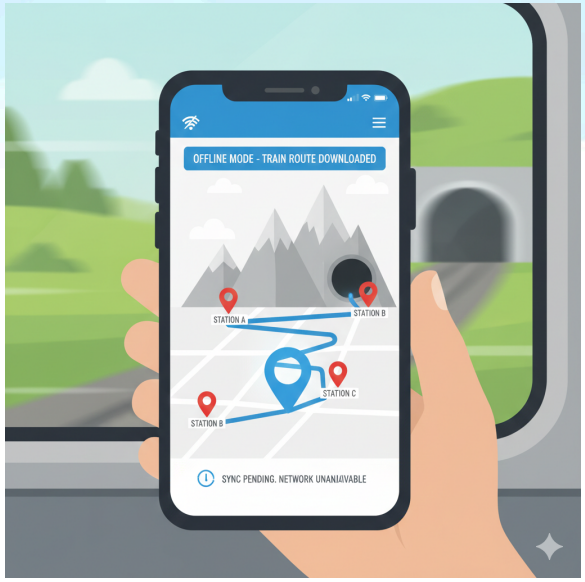
Just like Google Maps offline download:

Passenger downloads train route map before journey

- App stores:
- Train route path
- Station coordinates
- During journey:
- GPS works without internet
- Data syncs when network available

This avoids:

- Network dependency in tunnels
- Rural area connectivity issues



FLOWCHART (GPS Version)

Initialization: The process begins by **scanning a QR code**, which allows the app to automatically fetch the specific **Coach ID**.

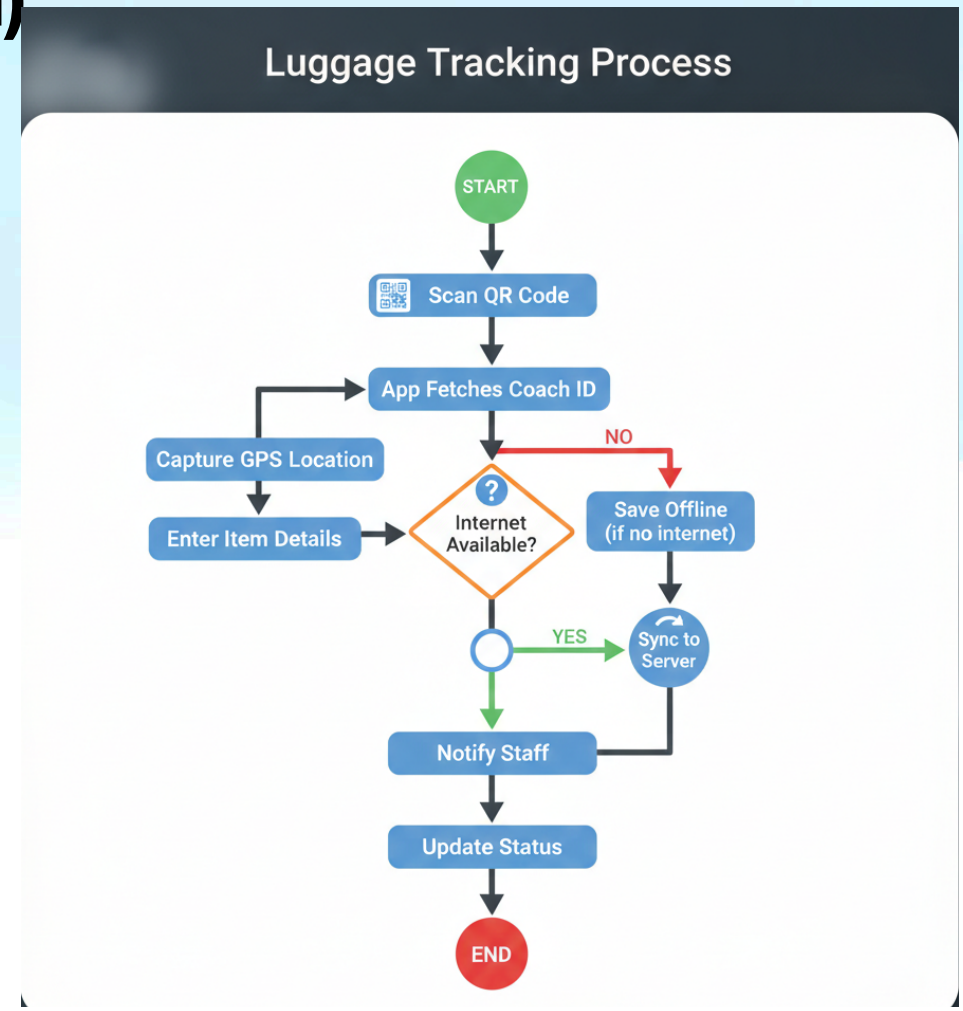
Data Collection: The system captures the current **GPS location** and prompts the user to manually **enter item details**.

Connectivity Check: A decision point determines if the internet is available; if not, the data is **saved offline** to the device.

Data Synchronization: Once a connection is established, the saved information is **synced to the server** to ensure record accuracy.

Communication: Upon successful sync, the system **notifies the staff** members responsible for the luggage.

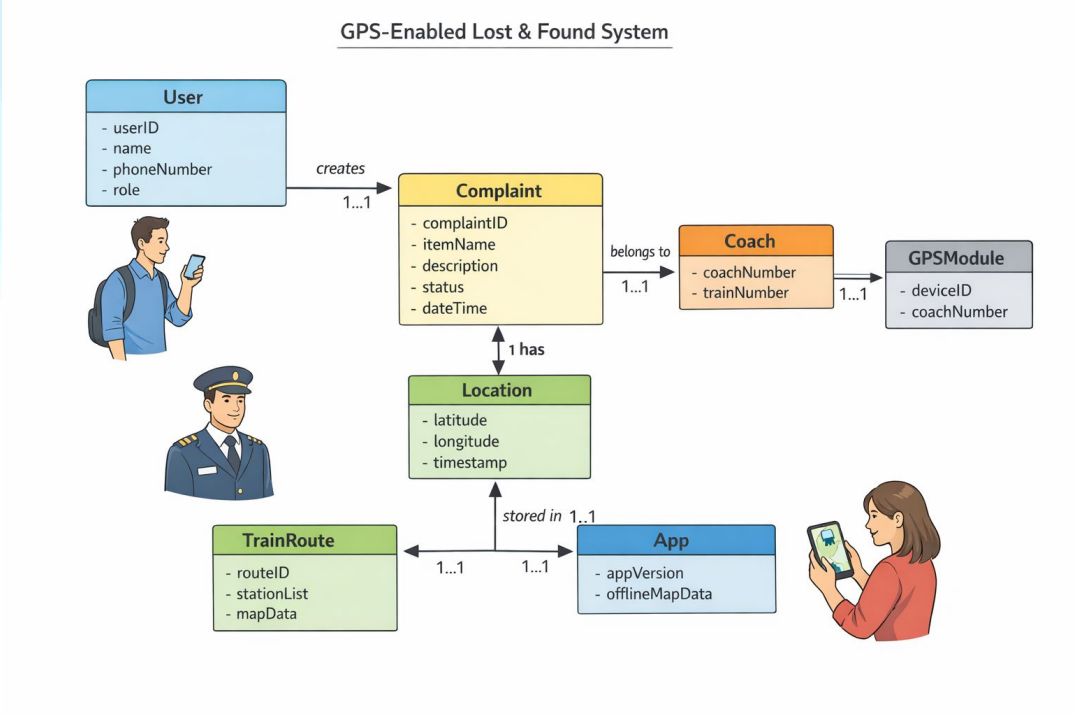
Finalization: The item's **status is updated** in the database, and the tracking workflow concludes.



UNITED MODELING LANGUAGE

Relationship Summary:

- One User → many Complaints
- One Complaint → one Location
- Many Complaints → one Coach
- One Coach → one GPSModule
- One TrainRoute → stored in App



BUDGET CONTROL

Instead of expensive hardware:

Option A (Low Budget):

•Use passenger mobile GPS only

No extra hardware

•Just QR code stickers in coach

Option B (Advanced):

•Install ₹1500–₹2000 GPS device per coach

SECURITY ASPECTS

- Location encryption
- Complaint ID hashing
- Role-based access
- Secure offline storage
- Data auto-delete after case closure

CONCLUSION

The GPS-Enabled Lost & Found System enhances passenger safety, transparency, and efficiency within Indian Railways by integrating real-time GPS tracking, offline route maps, and secure digital complaint management. The system ensures accurate location tagging, structured resolution processes, and strong data protection through encryption, role-based access, and automatic data deletion. By combining innovation with cost-effective implementation, this solution modernizes the traditional lost-and-found process, improves accountability, and significantly enhances the overall passenger experience.

THANK YOU